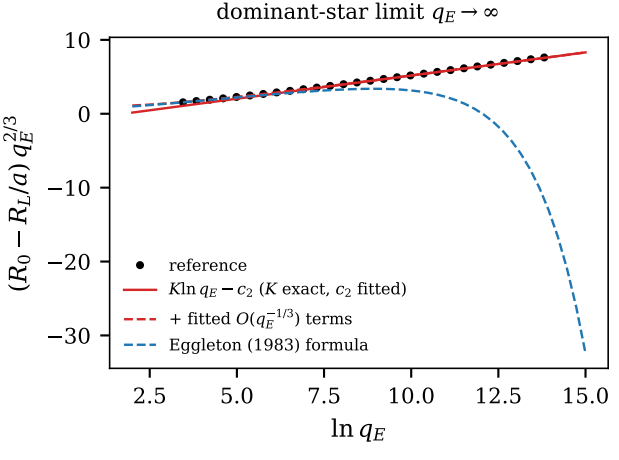
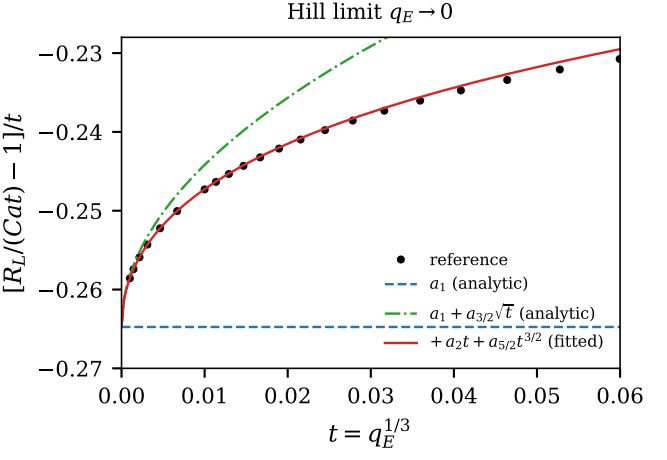

dominant-star limit $q_E \to \infty$
$(R_0 - R_L/a)\, q_E^{2/3}$
$\ln q_E$
reference
$K \ln q_E - c_2$ (K exact, c_2 fitted)
+ fitted $O(q_E^{-1/3})$ terms
Eggleton (1983) formula


Hill limit $q_E \to 0$
$[R_L/(Cat) - 1]/t$
$t = q_E^{1/3}$
reference
a_1 (analytic)
$a_1 + a_{3/2}\sqrt{t}$ (analytic)
$+ a_2 t + a_{5/2} t^{3/2}$ (fitted)